

Project Summary: Superstore Data Pipeline using Medallion Architecture

Project Title: End-to-End Data Pipeline on Azure

Objective:

Build a scalable data engineering pipeline that ingests, cleans, and aggregates raw Superstore sales data using **Azure Data Factory (ADF)** and **Azure Data Lake Storage Gen2 (ADLS Gen2)**, following **Medallion Architecture** (Bronze → Silver → Gold).

Key Components

- **Bronze Layer (Raw Data):**
Ingested CSV files into ADLS without transformation to maintain traceability.
 - **Silver Layer (Cleaned Data):**
Cleaned and transformed data using **ADF Mapping Data Flows**:
 - Fixed data types (Sales → Decimal, Quantity → Integer)
 - Filtered invalid rows
 - Added derived columns (YearMonth, ProfitMargin)
 - **Gold Layer (Aggregated Data):**
Aggregated datasets for business use:
 - Total Sales by date
 - Profit by Category
 - Revenue by Country
-

Technologies Used

- Azure Data Factory (ADF) – ETL pipelines and transformations
 - Azure Data Lake Storage Gen2 (ADLS Gen2) – layered data storage
 - Parquet File Format – columnar storage and performance
 - Data Pipeline Orchestration – modular and reusable pipelines
-

Achievements & Learnings

- Implemented **Medallion Architecture** in a real-world scenario.
 - Built **end-to-end, scalable ETL pipelines** in ADF.
 - Designed structured data storage with **Bronze/Silver/Gold layers**.
 - Applied **data validation, type conversions, and aggregations**.
 - Learned best practices for **data quality, partitioning, and schema evolution**.
-

Impact

- Delivered a **robust, production-ready data pipeline**.
- Generated **analytics-ready datasets** for downstream consumption.
- Demonstrated real-world **Data Engineering skills**: ingestion, transformation, storage, and pipeline orchestration.